

Lecture 5: Newton's Method: Local Convergence

Math 6610

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Today's Goals

- Extend Newton's method to systems via the Jacobian

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- Prove affine invariance and understand why it matters
- Preview the convergence theorem and the norms it needs

Newton's Method (Continued) I

Remark (*Newton's Method for $n \geq 1$*).

$$f : D \rightarrow \mathbb{R}^n, \quad f = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in D$$

Initial guess: $x^0 \in D$.

$$f(x) = f(x^0) + Df(x^0)(x - x^0) + O(\|x - x^0\|^2) \quad (1)$$

where

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} = \text{Euclidean norm.}$$

$$Df(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} = \left[\frac{\partial f_j}{\partial x_k} \right]_{j,k=1,2,\dots,n} = \text{Jacobian of } f$$

Newton's Method (Continued) II

Obtain a new approximation x^1 by setting (1) to 0:

$$0 = f(x^0) + \underbrace{Df(x^0) (x^1 - x^0)}_{\Delta x^0}$$

$$\begin{aligned} Df(x^0) \Delta x^0 &= -f(x^0) \\ x^1 &= x^0 + \Delta x^0. \end{aligned}$$

This is a system of n linear equations for n unknowns.

Remark (Newton's Method Algorithm). Choose $x^0 \in D$.

- for $k = 1, 2, \dots$ (open-ended), do:
 - if $Df(x^k)$ is singular, stop.
 - else, solve $Df(x^k) \Delta x^k = -f(x^k)$ for Δx^k
 - set $x^{k+1} = x^k + \Delta x^k$
 - if $x^{k+1} \notin D$, stop.
 - check for convergence: if $f(x^{k+1}) \approx 0$, stop.

Newton's Method (Continued) III

- end (k)

Remark.

- 1 Need to solve a linear system at each k step $\Rightarrow$ LU factorization of $A = Df(x^k)$
- 2 Newton's method is *affine-invariant*:
Let $M \in \mathbb{R}^{n \times n}$ be a fixed nonsingular matrix, and let $g : D \rightarrow \mathbb{R}^n$, $g(x) := Mf(x)$ for all $x \in D$. Thus, $f(x) = 0 \Leftrightarrow Mf(x) = 0 \Leftrightarrow g(x) = 0$. Then Newton's method applied to f and g results in the same iterates x^k , $k = 0, 1, 2, \dots$

Proof. (That Newton's method is affine-invariant).

$$g(x) = Mf(x)$$

$$Dg(x^k) = MDf(x^k)$$

$$Dg(x^k) \Delta x^k = -g(x^k) \Leftrightarrow Df(x^k) \Delta x^k = -f(x^k)$$

Convergence of Newton's Method I

Proposition (Convergence). Let x^* denote a zero of f , i.e. $f(x^*) = 0$. If $Df(x^*)$ is nonsingular and x^0 is “close” to x^* , then

$$\lim_{k \rightarrow \infty} x^k = x^*$$

and the speed of convergence is quadratic, i.e.

$$\|x^{k+1} - x^*\| \leq C \|x^k - x^*\|^2$$

Definition (Quadratic Convergence). The number of correct digits roughly doubles in each iteration.

Theorem (Convergence of Newton's Method, Affine-Invariant Version).

Assumptions:

- $D \subset \mathbb{R}$ is convex
- $f : D \rightarrow \mathbb{R}^n$ is C^1

Convergence of Newton's Method II

- $x^0 \in D$ and $Df(x^0)$ is nonsingular
- Df satisfies the affine-invariant Lipschitz condition:

$$\|(Df(x^0))^{-1}(Df(y) - Df(x))\| \leq \gamma \|y - x\|$$

Here $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ is a vector norm and $\|M\| := \max_{\|x\|=1} \|Mx\|$ is the associated matrix norm, which satisfies:

- 1 $\|Mx\| \leq \|M\| \|x\|$
- 2 $\|I\| = 1$
- 3 $\|MN\| \leq \|M\| \|N\|$

Filling in the Details: One Newton Step, Concretely I

Take the 2×2 system

$$f(x) = \begin{bmatrix} x_1^2 + x_2^2 - 1 \\ x_2 - x_1^2 \end{bmatrix} = 0 \quad (\text{circle} \cap \text{parabola}).$$

Jacobian:

$$Df(x) = \begin{bmatrix} 2x_1 & 2x_2 \\ -2x_1 & 1 \end{bmatrix}.$$

From $x^0 = (1, 1)^T$: $f(x^0) = (1, 0)^T$ and

$$\begin{bmatrix} 2 & 2 \\ -2 & 1 \end{bmatrix} \Delta x^0 = - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \Delta x^0 = \begin{bmatrix} -1/6 \\ -1/3 \end{bmatrix}, \quad x^1 = \begin{bmatrix} 5/6 \\ 2/3 \end{bmatrix}.$$

Note the workflow: evaluate f , evaluate Df , **LU-solve**, update. Never form $Df(x^k)^{-1}$ explicitly – solving the system is cheaper and more stable.

Filling in the Details: One Newton Step, Concretely II

Cost per Newton step for a dense Jacobian:

$$\underbrace{n^2 \text{ entries of } Df}_{\text{evaluation}} + \underbrace{\frac{2n^3}{3}}_{\text{LU}} + \underbrace{2n^2}_{\text{triangular solves}} .$$

The LU factorization dominates – this is why the linear algebra of Lectures 1–3 is the engine inside every nonlinear solver.

Why Affine Invariance Is the Right Point of View

Rescaling or mixing the equations, $g := Mf$ with M nonsingular, does not change the solution set – and the proof showed it does not change the Newton iterates either:

$$Dg(x^k)^{-1}g(x^k) = (MDf(x^k))^{-1}Mf(x^k) = Df(x^k)^{-1}f(x^k).$$

Consequences:

- Any *theory* for Newton's method should also be invariant under $f \mapsto Mf$. That is why the Lipschitz condition in the theorem is written with the factor $Df(x^0)^{-1}$ in front – the raw condition $\|Df(y) - Df(x)\| \leq \gamma \|y - x\|$ would change under M , but the affine-invariant one does not.
- Any *convergence test* should be invariant too. Checking $\|f(x^k)\|$ is not (multiply f by 10^6 !); checking the Newton correction $\|Df(x^k)^{-1}f(x^k)\| = \|\Delta x^k\|$ is. This foreshadows the monotonicity test of Lecture 7.

Looking Ahead

- **Next lecture:** The Newton Convergence Theorem